Fiscal Year:	FY 2024	**Task Last Updated:**	FY 09/28/2023
PI Name:	Gilroy, Simon Ph.D.		
Project Title:	Effect of Altered Gravity on the Nucleus		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Cell Culture		
PI Email:	sgilroy@wisc.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	608-262-4009
Organization Name:	University of Wisconsin-Madison		
PI Address 1:	Department of Botany		
PI Address 2:	430 Lincoln Dr.		
PI Web Page:			
City:	Madison	**State:**	WI
Zip Code:	53706-1313	**Congressional District:**	2
Comments:	NOTE: PI formerly at Pennsylvania State University; moved to University of Wisconsin-Madison in 2007 (Info received 7/2009)		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-MS, PS, AB. App D,E,F: Research Using Microgravity Simulation Devices, Parabolic and Suborbital Flights, and Antarctic Balloons
Start Date:	02/01/2020	**End Date:**	01/31/2023
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA KSC
Contact Monitor:	Zhang, Ye	**Contact Phone:**	321-861-3253
Contact Email:	Ye.Zhang-1@nasa.gov		
Flight Program:	Parabolic		
Flight Assignment:	NOTE: End date changed to 01/31/2023 per NSSC (Ed., 11/2/22)		
Key Personnel Changes/Previous PI:	NOTE: This project is a continuation of the investigation titled, "Effect of Altered Gravity on the Nucleus", with Dr. Simon Gilroy from the University of Wisconsin as the new Principal Investigator (PI). The original PI on the grant (Dr. Srujana Neelam) left the University of Wisconsin in April of 2021. (Ed., 2/2/23)		
COI Name (Institution):	Ullrich, Oliver M.D., Ph.D. (Universitat Zurich, Switzerland) Lele, Tanmay Ph.D. (University of Florida, Gainesville)		
Grant/Contract No.:	80NSSC20K0423		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: This project is a continuation with Principal Investigator (PI) Dr. Simon Gilroy at the University of Wisconsin. The previous PI was Dr. Dr. Srujana Neelam. Altered gravity is known to influence various cell functions like cell proliferation, signal transduction, and gene expression, in addition to the changes in cell morphology, focal adhesion, and cytoskeletal organization across a wide range of cell types. Such a wide range of phenotypic changes in altered gravity is linked to various abnormalities observed in astronauts returning from space but the biological mechanisms resulting in these changes are not yet understood. Our preliminary results suggest that simulated microgravity significantly alters nuclear morphology. Changes in nuclear shape can alter gene expression because the nucleus houses the genome, and changes in nuclear shape can alter chromatin conformation. Therefore, we propose to understand the mechanisms by which true altered gravity impacts nuclear morphology. Our hypothesis is that the LINC (Linker of Nucleoskeleton to Cytoskeleton) complex, a nuclear envelope complex which links the cytoskeleton to the nucleus, is sensitive to altered gravity mediated effects on the nuclear structure, nuclear tension, and subsequently on gene expression. We propose to identify the effects of altered gravity on nuclear morphology and cytoskeletal organization in human breast epithelial cells flown in parabolic flight. Cells flown will be cultured in a specially designed hardware to control for temperature and humidity, and the cells will be fixed on flight at different timepoints. High resolution imaging of cell shape and nuclear shape will be carried out using a laser scanning confocal microscope. Image analysis software will be used for three- dimensional shape reconstruction and quantification of shape parameters. We will characterize the mechanical forces acting on the nucleus using a Fluorescence Resonance Energy Transfer (FRET) probe biosensor to investigate the effect of altered gravity on the nuclear tension. Finally, we will identify the genes that are differentially expressed in LINC disrupted cells to understand the mechanotransduction pathway in sensing altered gravity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Cellular structures are permanently under the force of gravity in the Earth's gravitational field. Altered gravity has been demonstrated to have profound effects at the cellular and molecular level. Changes in cell morphology, proliferation, differentiation, signal transduction, and gene expression occur after exposure to altered gravity for just a few seconds. Each of the cytoskeletal elements and focal adhesion proteins become differentially expressed in altered gravity. Despite these results, it is not yet understood how change in gravitational forces alters the expression of genes. The LINC complex (Linker of Nucleoskeleton and Cytoskeleton) connects the nucleus and the surrounding cytoskeleton to mechanically couple the nuclear lamina with the cytoskeleton. Recent advances in our understanding of how nuclear envelope and nucleoskeleton associated elements contribute to both health and disease make it clear that cells require a functional mechanical coupling mediated by the LINC complex to respond normally to external mechanical forces. This research will answer the following question: Does altered gravity induce changes in force transmission through the LINC complex and LINC-associated gene expression by analyzing fibroblasts subjected to microgravity of parabolic flight?
Task Progress:	Altered gravity is known to influence various cell functions such as cell proliferation and morphology, focal adhesions, cytoskeletal organization, signal transduction, and gene expression across a wide range of cell types. Such extensive phenotypic changes to altered gravity may be linked to various abnormalities observed in astronauts returning from space, but the biological mechanisms resulting in these changes at the cellular level are not yet understood. The cytoskeleton is a major force bearing element in mammalian cells that is known to transduce forces both within and between cells. Cytoskeletal structure is also highly sensitive to the forces it receives. For example, each of the cytoskeletal elements and focal adhesion proteins become differentially expressed in altered gravity. The LINC complex (Linker of Nucleoskeleton and Cytoskeleton) connects the nucleus and the surrounding cytoskeleton to mechanically couple the nuclear lamina with the cytoskeleton. As such, it is a strong candidate for one element that might translate an altered gravity environment to changes in, e.g., gene expression. This research therefore sought to ask whether altered gravity induces changes in force transmission through the LINC complex and whether these effects play out as alterations in LINC-associated gene expression. Experimental Design: Altered gravity was delivered by using parabolic flight to impose periods of microgravity and hypergravity. These parabolic flights were conducted using human breast epithelial cells where the LINC complex was disrupted by expression of the construct SS-HA-SUN1L-KDEL. This protein contains part of the SUN1 protein. SUN proteins span the inner nuclear membrane and are integral elements of the LINC complex. SS-HA-SUN1L-KDEL is therefore a dominant-negative construct known to effectively disrupt the LINC complex. Identical cells expressing GFP-KDEL were used as a control for comparison between all treatments. In addition, cells expressing a Nesprin-FRET construct were also subjected to parabolic flight. Nesprin is another core element of the LINC complex of the nuclear envelope that faces the cytoplasm. A novel GFP-tagged Förtser Resonance Energy Transfer (FRET) construct of this protein anchors to the nuclear membrane but is predicted to change its FRET fluorescence signal as forces in the nucleus change. Despite some disruption in timing, personnel available to perform the parabolic flight experiments and travel by the pandemic, the parabolic experiments were successfully performed. • Flight #1 was performed by Novespace in Switzerland on 11th June 2020 with 15 parabolas. • The second flight was with Zero G in Fort Lauderdale on 13th November 2020 with 30 parabolas. • In both cases, the cell cultures were maintained in the Cell Culture Chamber hardware. The transfected samples were fixed at 1st and 7th parabolas and samples were successfully returned for subsequent analysis. Parallel ground controls were used to provide a 1xg comparison. Nuclear morphology and Nesprin-FRET sensor: Measurement of nuclear height and area showed no significant differences between any of the treatments. These observations suggest that despite employing highly sensitive confocal imaging to make the measurements, any relevant changes due to altered gravitational forces may have been below the resolution of the measurement approaches used. Similarly, the Nesprin sensor did not reveal any significant changes in FRET signal related to gravity environment. These sensor results were confirmed by an independent analysis of the data by the Conway group that developed the Nesprin tension sensor. Patterns of gene expression: Transcriptomics analysis however did suggest potential flight-related changes. RNA isolation from the samples harvested during parabolic flight was successful, with Illumina-based RNAseq being

	performed at the Interdisciplinary Center for Biotechnology Research (ICBR), University of Florida. Patterns of differential expression potentially related to phases of the parabolic flight were apparent in the LINC-disrupted cells. Application of the most stringent adjusted p-value to assign significance to changes in patterns of gene expression indicated variability in the samples. Such variability may be attributed to a batch effect on one set of replicates that appear as outliers. Therefore, further analyses were pursued using all 3 biological replicates from the experiment but with the $p<0.05$ cutoff for significance. From this analysis, the patterns of altered gene expression appear complex but microgravity and LINC-dependent changes are evident. For example, a cluster of genes are down-regulated in control cells experiencing the first parabola that are unaffected in the LINC-disrupted lines. Similarly, hypergravity affected a suite of genes in the LINC-disrupted cells that were unaffected in the controls. The response to the microgravity of the first parabola was not evident in the cells from the 7th parabola, suggesting rapid acclimation of cell response after the initial excursion into microgravity. Analysis was performed for gene set enrichment-by-treatment for components such as biological process, and molecular and cellular function. Thus, GeneSet Enrichment Analysis using the Human Molecular Signatures Database (Broad Institute and University of California San Diego / UCSD) revealed patterns likely linked to gravity response. For example, in the first parabola, LINC-disrupted cells showed up-regulation of a cluster of genes related to DSCAM (Down Syndrome Cell Adhesion Molecule) interactions. DSCAM is a regulator involved in a multitude of developmental events related to cell adhesion; whereas, under hypergravity, LINC-disrupted cells showed an apparent up-regulation of the reactome to RUNX2, a member of a transcription factor family involved in events such as skeletal morphogenesis and cell migration. Considering the cytoskeletal nature of the LINC complex, events related cell adhesion and migration are exciting predictions of potential roles. However, some unexpected patterns also appear, such as potential down-regulation of transport of organic cations.
Bibliography Type:	Description: (Last Updated: 02/02/2026)
Articles in Peer-reviewed Journals	Bakshi A, Swanson SJ, Gilroy S. "A touchy subject: Ca2+ signaling during leaf movements in Mimosa." Cell Calcium. 2023 Mar;110:102695. https://doi.org/10.1016/j.ceca.2023.102695 ; PMID: 36669253 , Mar-2023